1(A) Given: $\frac{d^2y}{dx^2} + \sin(x) \cdot \frac{dy}{dx} = \log(x)$ (1)

Function: $F(x, y, y', y'') = y'' + y' \cdot \sin x - \log(x)$

Domain: The logarithmic $\log(x)$ is defined for $x > 0$

$\therefore$ domain, $D = (0, \infty)$

1(B) Given: $\exp\left(\frac{d^3y}{dx^3} - \frac{dy}{dx}\right) + x \cdot \left(\frac{d^2y}{dx^2}\right)^2 = 0$

Function: $F(x, y, y', y'', y''') = \exp(y''' - y') + x(y'')^2$

Domain: Here, no explicit restriction on x from the functions involved (exponential & multiplication)

$\therefore D = \mathbb{R}$

1(c) Given: $\sum_{j=0}^{n} a_j \left(\frac{d^j y}{dx^j}\right) = 0$

Function: $F(x, y, y', y'', \ldots, y^{(n)}) = \sum_{j=0}^{n} a_j \, [y]^{(j)}$

Domain: It depends on function $a_j(x)$. As each $a_j(x)$ is defined on all of $\mathbb{R}$. $\Rightarrow D = \mathbb{R}$.

(2) Given $\frac{dy}{dx} = \sqrt{|y|}$ with $y(0) = 0$

Case-1: $y \geq 0$. Then $\sqrt{|y|} = \sqrt{y} \Rightarrow \frac{dy}{\sqrt{y}} = dx$

$\therefore 2\sqrt{y} = x + C \Rightarrow y = \frac{(x+C)^2}{4}$

Applying $y(0) = 0$; we get $0 = \frac{C^2}{4} \Rightarrow C = 0$

$\therefore$ One solution is: $\boxed{y_1(x) = \frac{x^2}{4}}$

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Case-2: $y < 0$: Then $\sqrt{|y|} = \sqrt{-y}$.

$\Rightarrow \frac{dy}{\sqrt{-y}} = dx$. Take $y = -v$, where $v > 0$:

$\Rightarrow \frac{d(-v)}{\sqrt{v}} = dx \Rightarrow -2\sqrt{v} = x + C$

$\therefore y = -\frac{(x+C)^2}{4}$.

Applying $y(0) = 0$, $\Rightarrow 0 = \frac{C^2}{4} \Rightarrow C = 0$

$\therefore$ Another solution is: $\boxed{y_2(x) = \frac{-x^2}{4}}$

$\Rightarrow$ Given $y(0) = 0$, another trivial solution is $y(x) = 0$ for all x, since $\sqrt{|0|} = 0$.

$\therefore$ Third solution is: $\boxed{y_3(x) = 0}$

$\Rightarrow$ Constructing 4^{th} Solution: To construct a piecewise solution:

$\rightarrow$ For $x \leq 0$: We choose solⁿ $y_3(x) = 0$. This choice satisfies the DE & initial condition $y(0) = 0$

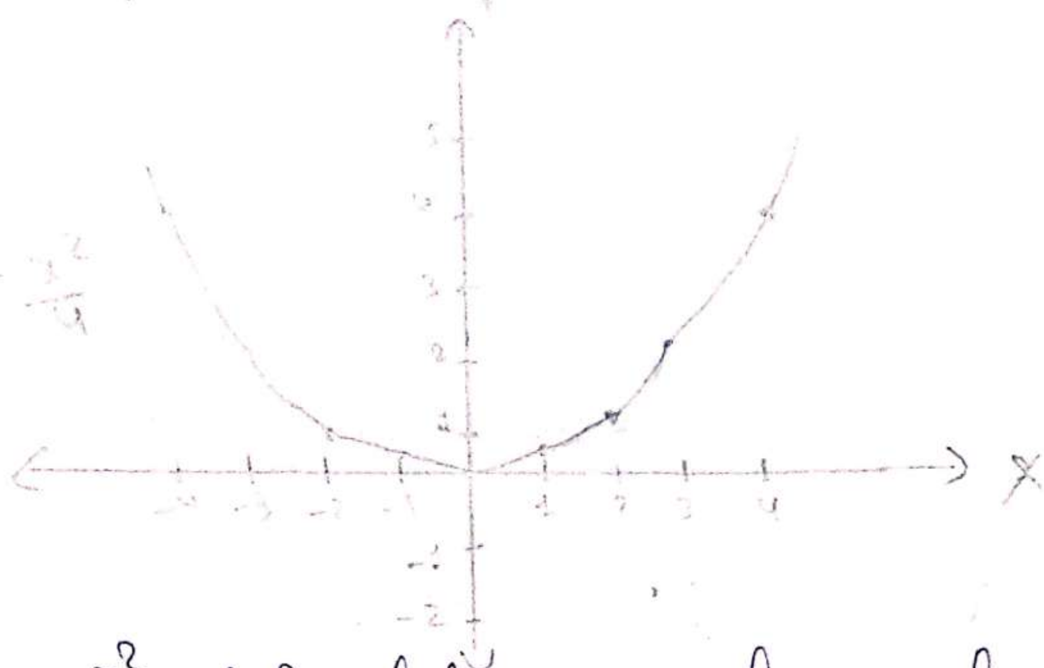
$\rightarrow$ For $x > 0$: We can switch to another solution. A natural choice is $y_1(x) = x^2/4$, which also satisfies the DE & initial condition $y(0) = 0$.

~~This~~ Thus, we can define a new piecewise function, that is the fourth solution as:

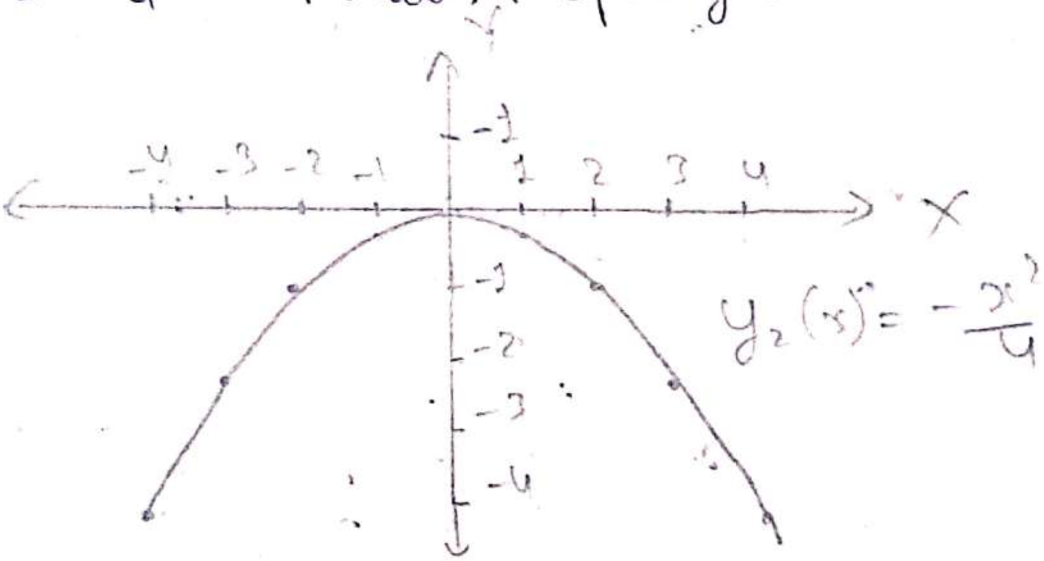
$$\boxed{y_4(x) = \begin{cases} 0 ; & x \leq 0 \\ \frac{x^2}{4} ; & x > 0 \end{cases}}$$

Sketch the Solution

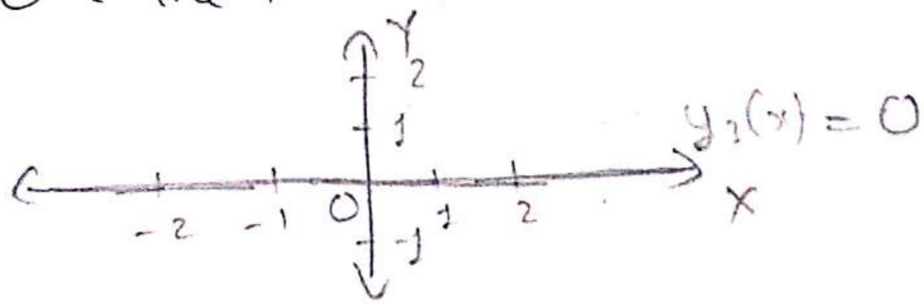
→ $y_1(x) = \frac{x^2}{4}$: A parabola opening upwards.



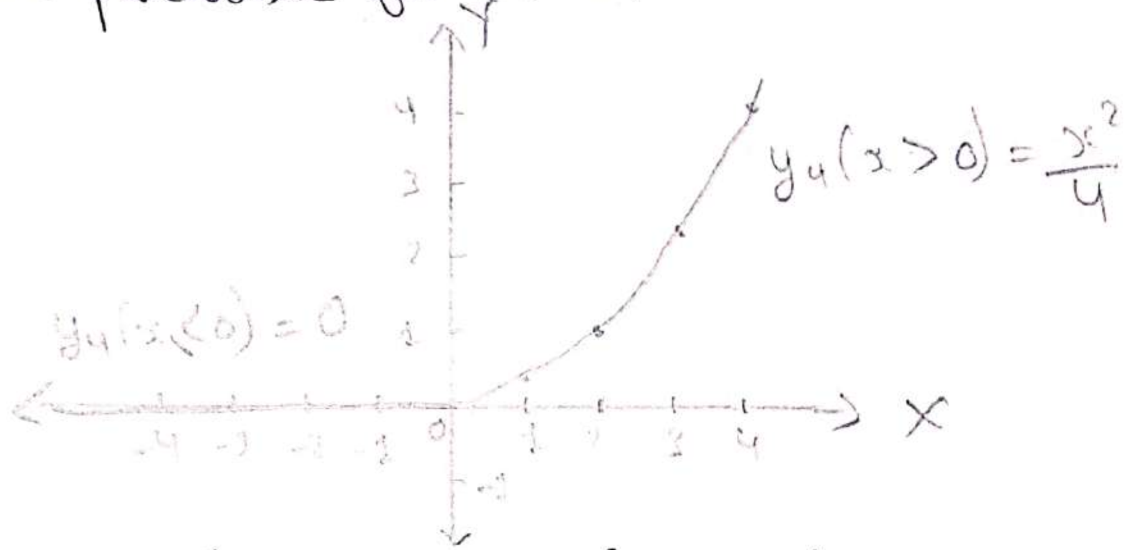
→ $y_2(x) = -\frac{x^2}{4}$: A Parabola opening downwards



→ $y_3(x) = 0$: The X-axis



$\rightarrow y_4(x)$: A piecewise function, flat on left of $x = 0$ & then parabolic for $x > 0$. (4)



$\Rightarrow$ Well-Posedness: The well-posedness of a problem typically require that a solution exists, a unique & depends continuously on the initial conditions. Here:

$\rightarrow$ Existence: Solutions do exist, as shown above

$\rightarrow$ Uniqueness: The solution is not unique, as we have four different solutions.

$\rightarrow$ Continuous Dependence: This is violated due to non-uniqueness, indicating that small changes in initial conditions could yeild dramatically diff^n solution.

$\therefore$ Initial value problem is not well-posed due to the lack of uniqueness.

(3) Step-1: Solve for $0 \leq x \leq 1$:

Here, equ^n becomes: $\frac{d}{dx} y(x) + \lambda_1 y(x) = \lambda_3$

This is standard first-order linear differential equati

$\therefore$ Integrating factor, $\mu(x) = e^{\int \lambda_1 dx} = e^{\lambda_1 x}$

$\therefore e^{\lambda_1 x} \frac{d}{dx} y(x) + \lambda_1 e^{\lambda_1 x} y(x) = \lambda_3 . e^{\lambda_1 x}$

$\therefore \frac{d}{dx} [e^{\lambda_1 x} . y(x)] = \lambda_3 e^{\lambda_1 x}$. Integrating both sides

$\Rightarrow e^{\lambda_1 x} y(x) = \frac{\lambda_3}{\lambda_1} e^{\lambda_1 x} + C_1,$

where, C_1 is constant of integration

$\therefore$ Solution for $y(x)$ in region $0 \leq x \leq 1$ is :-

$\cancel{y(x) = \frac{\lambda_3 e^{\lambda_1 x}}{\lambda_1} + C_1}$

$$\boxed{y(x) = \frac{\lambda_3}{\lambda_1} + C_1 e^{-\lambda_1 x}}$$

Step-2: Solve for $x > 1$:

Here, equ^n becomes: $\frac{d}{dx} y(x) + \lambda_2 y(x) = \lambda_4$

$\therefore$ Integrating factor $= e^{\int \lambda_2 dx} = e^{\lambda_2 x}$

$\therefore e^{\lambda_2 x} \frac{d}{dx} y(x) + \lambda_2 e^{\lambda_2 x} y(x) = \lambda_4 e^{\lambda_2 x}$

$\therefore \frac{d}{dx} [e^{\lambda_2 x} . y(x)] = \lambda_4 e^{\lambda_2 x}$. Integrating, we get

$\Rightarrow e^{\lambda_2 x} . y(x) = \frac{\lambda_4}{\lambda_2} e^{\lambda_2 x} + C_2$

where C_2 is another constant of integration

$\therefore$ Solution for $y(x)$ in region $x > 1$ is:

$$\boxed{y(x) = \frac{\lambda_4}{\lambda_2} + C_2 e^{-\lambda_2 x}}$$

Step-3: Continuity condition at $x = 1$:

To ensure the solution is continuous at $x = 1$, we must have: $y(1^-) = y(1^+)$

Substituting $x = 1$ into sol^n from both regions:

$$\frac{\lambda_3}{\lambda_1} + C_1 e^{-\lambda_1} = \frac{\lambda_4}{\lambda_2} + C_2 e^{-\lambda_2}$$

This equation allows us to solve for constants C_1 & C_2 given initial or boundary conditions.

Step-4: General Solution

The continuous solution to differential equ^n is given by:

$$y(x) = \begin{cases} \frac{\lambda_3}{\lambda_1} + C_1 e^{-\lambda_1 x}; & \text{if } 0 \leq x \leq 1 \\ \frac{\lambda_4}{\lambda_2} + C_2 e^{-\lambda_2 x}; & \text{if } x > 1 \end{cases}$$

where, C_1 & C_2 are constants determined by initial condi or the continuity condition at $x = 1$.

4(a) Given $\left(\frac{1}{x}-\frac{1}{y}\right)dx + \left(\frac{x}{y^2}\right)dy = 0$

Compare with: $M(x,y)\,dx + N(x,y)\,dy = 0$

we get, $M(x,y) = \frac{1}{x} - \frac{1}{y}$ and $N(x,y) = \frac{x}{y^2}$

$\Rightarrow \frac{\partial M}{\partial y} = 0 - \left(-\frac{1}{y^2}\right) = \frac{1}{y^2}$

$\Rightarrow \frac{\partial N}{\partial x} = \frac{1}{y^2}$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow$ the equation is Exact.

To solve this, we find a function $\psi(x,y)$ such that:

$\frac{\partial \psi}{\partial x} = M(x,y)$ and $\frac{\partial \psi}{\partial y} = N(x,y)$

$\Rightarrow$ Integrate $M(x,y)$ w.r.t. x, we get:

$\psi(x,y) = \int \left(\frac{1}{x} - \frac{1}{y}\right)dx = \ln|x| - \frac{x}{y} + h(y)$

where, $h(y)$ is arbitrary function of y.

$\Rightarrow$ Differentiate $\psi(x,y)$ w.r.t. y, we get

$\frac{\partial \psi(x,y)}{\partial y} = +\frac{x}{y^2} + h'(y) = N(x,y)$

$\therefore +\frac{x}{y^2} + h'(y) = \frac{x}{y^2}$

$\therefore h'(y) = 0 \Rightarrow h(y)$ is constant.

$\therefore$ Solution is: $\boxed{\ln|x| - \frac{x}{y} = C}$

where, C is constant.

tion

4(B) Given:

$[y\cos(x) - \sin(y)]dx + [\sin(x) - x\cos(y)]dy = 0$

Here, $M(x,y) = y\cos(x) - \sin(y)$ & $N(x,y) = \sin(x) - x\cos(y)$

$\Rightarrow \frac{\partial M}{\partial y} = \cos x - \cos y$ & $\frac{\partial N}{\partial x} = \cos x - \cos y$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow$ Equation is exact.

$\Rightarrow$ To solve this, we find a function $\psi(x,y)$ such that:

$\frac{\partial \psi}{\partial x} = M(x,y)$ & $\frac{\partial \psi}{\partial y} = N(x,y)$

$\Rightarrow$ Integrate $M(x,y)$ w.r.t., x, we get

$\psi(x,y) = \int [y\cos x - \sin y]dx = y\sin x - x\sin y + g(y)$

$\Rightarrow$ Differentiate $\psi(x,y)$, w.r.t., y:

$\frac{\partial \psi(x,y)}{\partial y} = \sin x - x\cos y + g'(y) \stackrel{?}{=} N(x,y)$

$\therefore \sin x - x\cos y + g'(y) = \sin x - x\cos y$

$\therefore g'(y) = 0 \Rightarrow g(y)$ is constant.

Thus, the solution is:

$\boxed{y\sin(x) - x\sin(y) = C}$, where C is constant.

Exact.

4 (c) Given

$$(1-xy)^{-2}dx + [y^2+x^2(1-xy)^{-2}]dy = 0$$

Here, $M(x,y) = (1-xy)^{-2}$ and $N(x,y) = [y^2+x^2(1-xy)^{-2}]$

$\Rightarrow \frac{\partial M}{\partial y} = 2x(1-xy)^{-3}$ and $\frac{\partial N}{\partial x} = 2x(1-xy)^{-2} - 2x^2y(1-xy)^{-3}$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \Rightarrow$ The equation is NOT exact.

(5)

Part - 1: To show: $\phi(x+p)$ is also a solution.

$\Rightarrow$ Differentiating $\phi(x+p)$, w.r.t., x:

$$\frac{d}{dx}[\phi(x+p)] = \frac{d\phi}{d(x+p)} \cdot \frac{d(x+p)}{dx} = \frac{d\phi}{d(x+p)} \cdot 1 = \frac{d\phi}{dx}(x+p)$$

Substitute $y(x) = \phi(x)$ into differential equation:

$$\therefore \frac{d\phi(x)}{dx} + f(x)\phi(x) = 0$$

~~Put~~ Substitute $x+p$ for x in above; we get

$$\frac{d}{dx}[\phi(x+p)] + f(x+p)\,\phi(x+p) = 0$$

$$\therefore \frac{d\phi}{dx}(x+p) + f(x+p)\,\phi(x+p) = 0$$

Since $f(x)$ is periodic with period p, we have,
$f(x+p) = f(x)$.

$$\therefore \frac{d\phi}{dx}(x+p) + \cancel{f(x+p)}\, f(x)\,\phi(x+p) = 0$$

Hence, $\phi(x+p)$ is also a solution of differential equation.

Part-2: To show: $\phi(x+p) = c\cdot\phi(x)$

We know, $\phi(x)$ & $\phi(x+p)$ both are solution of given differential equation

$\Rightarrow$ consider the function, $\psi(x) = \dfrac{\phi(x+p)}{\phi(x)}$.

To show: $\psi(x)$ is constant, i.e., $\psi'(x) = 0$

$\Rightarrow$ Differentiate $\psi(x)$, w.r.t., x:

$$\psi'(x) = \frac{\phi'(x+p)\,\phi(x) - \phi(x+p)\,\phi'(x)}{[\phi(x)]^2}$$

Since $\phi(x+p)$ is solution of given differential equation:

$$\therefore \phi'(x+p) = -f(x+p)\,\phi(x+p)$$

Also, $\phi'(x) = -f(x)\,\phi(x)$.

$$\therefore \psi'(x) = \frac{-f(x+p)\phi(x+p)\phi(x) - \phi(x+p)[-f(x)\cdot\phi(x)]}{[\phi(x)]^2}$$

$$\therefore \psi'(x) = \frac{-f(x)\phi(x+p)\phi(x) + \phi(x+p)f(x)\phi(x)}{[\phi(x)]^2} \quad [\because f(x+p) = f(x)]$$

$\therefore \psi'(x) = 0 \Rightarrow \psi(x)$ is constant

$\therefore \dfrac{\phi(x+p)}{\phi(x)} = c$, where c is some constant

$$\therefore \boxed{\phi(x+p) = c\cdot\phi(x)}$$

This completes the proof.